Learning From Human Insights for Remote Sensing Change Detection

Zhihao Li, Yikun Liu, Minghao Liu, and Gongping Yang

***Abstract*—Remote sensing change detection (RSCD) aims to accurately identify changed objects within the same region over time, facilitating the interpretation of Earth's surface dynamics. However, RSCD faces two fundamental challenges: the variation in partial representations of identical objects over time and the similarity in certain properties of different objects, both of which significantly hinder performance. Drawing inspiration from the human difference analysis strategy, which detects changes by empirically selecting and integrating interobject comparisons across various properties (such as color, boundary, and texture), a difference analyzer (DA) is introduced. The DA identifies changes based on property-based features rather than holistic spatial features, which enables adaptive suppression of inconsistent properties of the identical object over time to eliminate pseudo changes, as well as the suppression of similar properties of different objects to accurately identify the changes, thus effectively overcoming the aforementioned challenges. Specifically, the bitemporal property descriptors are derived using a newly proposed sliding discrete cosine transform (SDCT), where distinct channels represent various object properties. Subsequently, a cross-descriptor attention (CDA) is developed that selectively activates discriminative properties (such as building boundaries) while suppressing interfering ones (such as illumination-induced color variations), enabling effective selected property integration and comparisons. Next, the DA constructs property difference descriptors that represent the interobject comparisons. To embed DA within an end-to-end network, a learning from human insights for change detection network (LHICD-Net) is proposed, which replicates the human RSCD process through perception, analysis, refinement, and prediction stages. To match human perception in diverse bands, a high-frequency residual structure (HRS) that can preserve high-frequency information during downsampling is incorporated in the encoder. A refinement module (RM) is also designed to enhance spatial details by extracting fine-grained features from diverse receptive fields, corresponding to the human refinement stage. Importantly, we reveal that the working mechanism of the DA aligns with the human difference analysis strategy and validate the superiority of our property descriptor generation method. Extensive experiments confirm that LHICD-Net outperforms state-of-the-art methods across three widely used RSCD datasets. The source code is available at https://github.com/yihui1230/LHICDNet**

***Index Terms*—Change detection (CD), frequency learning, human difference strategy, property-based comparison.**

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I. Introduction

REMOTE sensing change detection (RSCD), serves as a fundamental task in the analysis of remote sensing (RS) imagery and has been widely utilized across various domains, such as urban development assessment [1], land cover change monitoring [2], and earthquake damage evaluation [3].

Traditional RSCD methods primarily depend on prior knowledge and empirical designs, encompassing algebra-based [6], transformation-based [7], and classification-based approaches [8]. These methods often suffer from limited generalizability and inadequate modeling capabilities in complex RS scenarios.

With the rapid progress in computer vision model development [9] and the enhanced quality of RS data [10], deep learning-based Siamese networks [11], [12], [13], [14] have recently gained prominence in RSCD due to their improved performance. Typically, these methods employ Siamese network architectures [15] to independently extract spatial features from bitemporal images, followed by an analysis of the spatiotemporal relationships between these features to identify changes. Most existing approaches share a common objective: to design more effective temporal interactions between bitemporal spatial features for accurately modeling their relationships. Current temporal interaction strategies in RSCD generally fall into four categories: elementwise subtraction [12], channelwise concatenation [16], channelwise/spatialwise exchange [17], and attention-based fusion [18], [19], all of which facilitate the model's ability to recognize commonalities and differences between bitemporal images. These interaction strategies can be interpreted as the model's strategy for difference analysis.

Existing methods, driven by their respective difference analysis strategies, have demonstrated remarkable performance. Nevertheless, certain issues remain unresolved, including pseudo and unidentified changes, as illustrated in Fig. 1(b). These limitations can be attributed to two fundamental challenges in RSCD: the variation in partial representations of identical objects over time and the similarity in certain properties of different objects.

In contrast to existing methods, it is observed that humans can easily and accurately identify actual changes. This fact motivates a re-examination of research on human change detection (CD). For example, Rensink [5] reported that human performance in detecting orientation changes remains largely unaffected by distracting contrast variations. In addition,

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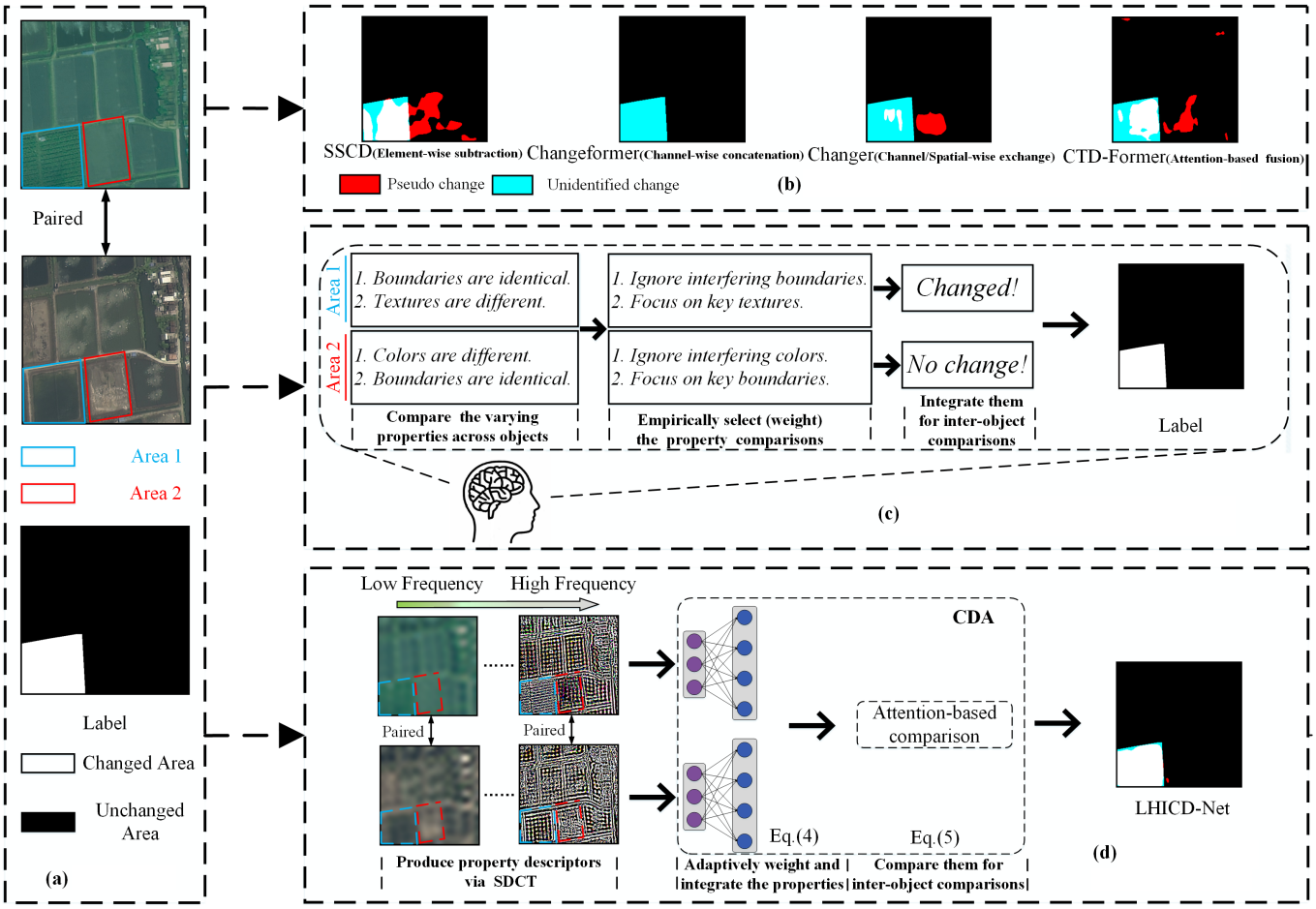


Fig. 1. (a) Two fundamental challenges in RSCD: variation in the partial representations of identical objects over time and similarities in some of the properties of different objects; these cause pseudo and unidentified changes, respectively. (b) Existing RSCD approaches, whose different analysis strategies are based on spatial feature comparisons, struggle to address these issues effectively. (c) In contrast, humans, using their different analysis strategies [4], [5], can reliably overcome these challenges. (d) Overview of our proposed difference analysis strategy, realized through the DA. While the execution sequence differs, the DA successfully emulates the human difference analysis process. It is worth noting that low-frequency features primarily encode color properties, whereas high-frequency features predominantly represent boundary and texture properties.

Rensink [4] demonstrated that humans are adept at concentrating on the key properties of changing objects while disregarding irrelevant ones. These studies suggest that the human strategy involves comparing objects based on specific, varying properties rather than considering them as complete entities. The key principle behind this strategy is that interfering and discriminatory property comparisons can be separated and selected adaptively, thereby addressing the aforementioned challenges. In addition, we outlined a simplified description of the human RSCD process, which can be broadly categorized into four stages: 1) perception stage, where humans first perceive objects along with their varying properties; 2) analysis stage, as illustrated in Fig. 1(c), where humans compare objects in bitemporal images based on their varying properties and empirically select these property comparisons to obtain interobject comparisons; 3) refinement stage, where after identifying approximate change regions through the interobject comparisons, humans further refine the details of these regions [20]; and 4) prediction stage: where precise pixel-level predictions are ultimately made.

In response to the above, we propose the learning from human insights for CD network (LHICD-Net), an end-to-end framework designed to emulate these four stages of the human RSCD process. Specifically, in the perception stage, the encoder is used to perceive objects and their varying properties by transforming bitemporal images into model-recognizable deep features. A high-frequency residual structure (HRS) is incorporated to preserve high-frequency details, such as textures. Furthermore, in the analysis stage, a difference analyzer (DA) is constructed to implement the human difference analysis strategy. Using the proposed sliding discrete cosine transform (SDCT), the bitemporal features are decomposed into different frequency bands, generating property descriptors with customized spatial sizes and band lengths. Following the rearrangement method in [21], each channel of these descriptors corresponds to spatial features within a specific band, thus representing distinct object properties. The customized design of SDCT makes these descriptors more suitable for pixel-level classification in RSCD. A cross-descriptor attention (CDA), inspired by [19], is introduced to selectively activate

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discriminative properties and suppress interfering ones, facilitating effective selected property integration and comparison. Next, the DA produces property difference descriptors that represent the interobject comparisons. Benefiting from DA, our method achieves effective disentanglement of bitemporal features into distinct properties, which enables selective activation of discriminative properties and suppression of interference properties, thus overcoming the challenges described in Fig. 1(a). As shown in Fig. 1, when the identical cropland (in red box) captured under different light conditions exhibits varying color properties and identical boundary properties, our DA first decouples the bitemporal features into frequency-band-specific properties, such as color properties in low-frequency bands and boundary properties in high-frequency bands. By leveraging learnable linear layers in CDA, our method selectively activates discriminative boundary properties while suppressing interfering color properties, thereby effectively eliminating the pseudo changes. Similarly, when different croplands (in blue box) share similar boundary properties, the DA can adaptively suppress the interfering boundary properties and enhance discriminative texture properties, enabling accurate identification of changed cropland. In the refinement stage, these property difference descriptors are then mapped to a finer resolution using a cross-transformer decoder. In addition, a refinement module (RM) is employed to further enhance spatial details by refining the input features. Finally, in the prediction stage, the RSCD prediction maps are generated by the predictor.

The primary contributions of this article are outlined as follows.

1) Drawing inspiration from the human difference analysis strategy, which focuses on property-based comparisons, we introduce the DA to implement this approach. DA generates property descriptors using SDCT, adaptively selects relevant properties through CDA, and derives interobject comparisons based on the selected property comparisons.
2) To effectively incorporate DA within a deep learning framework, we propose an end-to-end network named LHICD-Net, which faithfully replicates the four stages of the human RSCD process. The encoder, integrated with HRS, functions as the perception stage, which preserves high-frequency properties to align with human perception across diverse bands. DA is responsible for the analysis stage. The cross-transformer decoder and RM together constitute the refinement stage by extracting fine-grained features from diverse receptive fields. The prediction stage is realized by the predictor.
3) We provide a deeper insight into the internal mechanism of our DA, demonstrating its consistency with the human difference analysis strategy through the investigation of property selection. The effectiveness of our property-based comparison is further validated in the ablation study by evaluating the property descriptor generation methods.
4) Extensive experiments conducted on three RSCD benchmark datasets show that LHICD-Net exhibits superior performance and consistently outperforms existing state-of-the-art methods.

II. Related Work

A. Frequency Learning

In comparison to commonly used spatial features, frequency features offer more informative cues, despite the introduction of some noise. More importantly, frequency features enable representation decomposition, wherein each band corresponds to a distinct property. This decomposition capability has made frequency learning particularly effective in image-level tasks, such as denoising [22], deblurring [23], and pan-sharpening tasks [24]. An increasing number of studies [25], [26], [27], [28] have focused on frequency learning, contributing both to enhanced model generalizability [29], [30], [31] and improved performance [32], [33], [34]. For pixel-level tasks, Xu et al. [21] were the first to use processed frequency features as the input to a deep learning network instead of raw images, demonstrating their effectiveness in segmentation. This work subsequently inspired broader interest in learning within the frequency domain. Zhou et al. [35] proposed a joint frequency-spatial domain approach that leveraged rich frequency information to complement spatial features, highlighting the potential of frequency utilization in RSCD. However, these methods typically rely on conventional discrete cosine transform (DCT) [21], which are constrained by fixed spatial sizes and rigid partitioning strategies. As a result, the extracted spatial–frequency features are poorly suited for pixel-level classification tasks.

Alternatively, Yao et al. [36] employed the discrete wavelet transform (DWT) [37] within a transformer framework to balance efficiency and accuracy. Nevertheless, it is argued that the four-band output of DWT may not sufficiently capture the complexity present in RS scenarios. In summary, current frequency feature extraction methods do not fully utilize the decomposition potential offered by frequency learning. To address this limitation, we introduce SDCT, which allows for customized spatial sizes and band lengths, thereby providing a high-quality foundation for subsequent difference analysis.

B. Learning From Humans and CD

From early models, such as perceptrons [38] to more advanced architectures such as long short-term memory (LSTM) [39] and spiking neural networks (SNNs) [40], many computational approaches have been inspired by human cognition with the goal of matching or even surpassing human capabilities. The success of these models indicates that learning from humans is often a valid strategy, particularly for tasks that humans can easily perform. For instance, Zhong et al. [41] enhanced camouflage object detection by incorporating frequency-domain features as an additional cue beyond human visual capabilities. Xu et al. [42] proposed a robust SNN that inputs image data at different frequency bands over time, mimicking the selective visual attention mechanism found in biological systems to resist noise interference.

Research, such as Desimone et al. [43] and Zani and Proverbio [44], has also emphasized that selective visual

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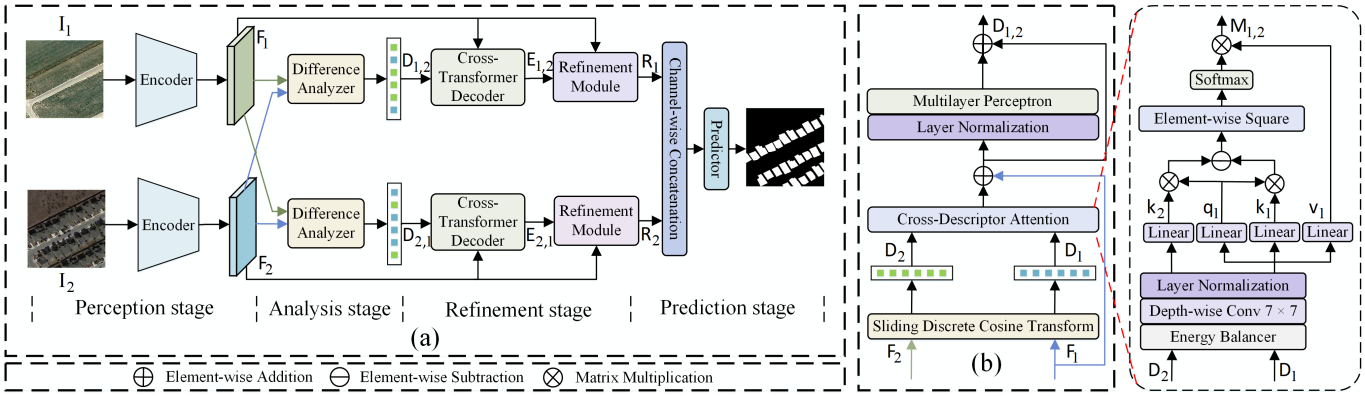


Fig. 2. (a) Overview of the proposed LHICD-Net, which effectively models the four stages of the human RSCD process. (b) Illustration of the proposed DA, using the first instance as an example. For clarity, reshape operations are omitted, and a single-head CDA mechanism is depicted.

attention allows the human visual system to concentrate on information from specific frequency bands while filtering out irrelevant signals. This mechanism is equally important for RSCD, as it supports the property selection process fundamental to the human difference analysis strategy. Sun et al. [45] proposed a structure consistency-based method for RSCD, which identified changes by selectively comparing the structural features of bitemporal images, rather than comparing components that are sensitive to noise, light variations, and other interfering factors. Similarly, Sun et al. [46] proposed a locality-preserving energy model to selectively enforce the structural consistency constraint.

Recently, standard difference analysis strategies in RSCD, including elementwise subtraction, channelwise concatenation, channelwise/spatialwise exchange, and attention-based fusion, remain widely used. Although recent RSCD methods employing these strategies [12], [13], [14], [19], [47], [48], [49], [50], [51] have achieved notable progress, we argue that they may be suboptimal, as they do not align with the human difference analysis strategy. Furthermore, the persistent issues of pseudo and unidentified changes provide additional evidence supporting this argument.

III. Method

The core concept of the proposed LHICD-Net is to emulate and incorporate the human difference analysis strategy within a deep learning framework, with the objective of advancing RSCD.

A. Overview

As shown in Fig. 2(a), the LHICD-Net workflow is composed of four stages: perception, analysis, refinement, and prediction. In the perception stage, the bitemporal images $\{I_1, I_2\} \in \mathbb{R}^{3\times H\times W}$ are processed by the encoder, which extracts model-recognizable deep features $\{F_1$ and $F_2\} \in \mathbb{R}^{C\times H/4\times W/4}$ (where C denotes the number of channels).

In the analysis stage, property descriptors $\{D_1, D_2\}$ and property difference descriptors $\{D_{1,2}, D_{2,1}\}$ are generated successively by DA, each in the space $\in \mathbb{R}^{C\times p\times p\times H/(4\times s)\times W/(4\times s)}$ (where $p \times p$ and s denote the SDCT window size and stride), thereby, effectively implementing the human difference analysis.

Next, in the refinement stage, the property difference descriptors are mapped into a more fine-grained resolution by a cross-transformer decoder to yield $\{E_{1,2}, E_{2,1}\} \in \mathbb{R}^{C\times H/4\times W/4}$. In addition, the RM enhances spatial detail by extracting fine-grained features across diverse receptive fields, forming refined features $\{R_1, R_2\} \in \mathbb{R}^{C_R\times H/4\times W/4}$ (with C_R representing the refined channel dimension).

Finally, in the prediction stage, the channelwise concatenation of R_1 and R_2 is passed to the predictor to generate the final output maps.

B. Encoder

In LHICD-Net, we employ the widely adopted ResNet-18 [9], as the encoder to extract model-recognizable features from bitemporal input images. To better match human perception across multiple frequency bands, we introduce the HRS to mitigate and reduce high-frequency information loss during downsampling. Specifically, given a 2-D feature map X, we apply the DWT to decompose X into four subbands: X_{HH}, X_{HL}, X_{LH}, and X_{LL}, where X_{LL} refers to the low-frequency component, while X_{HH}, X_{HL}, and X_{LH} correspond to the high-frequency components. The high-frequency residual $\overline{X}$ is constructed from these components and added to the downsampled feature map, thereby preserving critical texture and edge information. The HRS process is defined as follows:

$$\begin{aligned} X_{HH}, X_{HL}, X_{LH}, X_{LL} &= \mathrm{DWT}(X) \\ \overline{X} &= \mathrm{Conv}_{1\times 1}(\mathrm{Cat}(X_{HH}, X_{HL}, X_{LH})) \\ Y &= \mathrm{Downsample}(X) + \overline{X} \end{aligned} \tag{1}$$

where $\mathrm{Conv}_{1\times1}(\cdot)$ denotes a 1×1 convolution, $\mathrm{Cat}(\cdot)$ denotes channelwise concatenation, and $\mathrm{Down-sample}(\cdot)$ is implemented by a 3×3 convolution with a stride of 2.

C. Difference Analyzer

As illustrated in Fig. 2(b), the DA is designed to implement the core difference analysis strategy effectively. Specifically, DA first employs the SDCT to decompose object properties and produce the corresponding property descriptors

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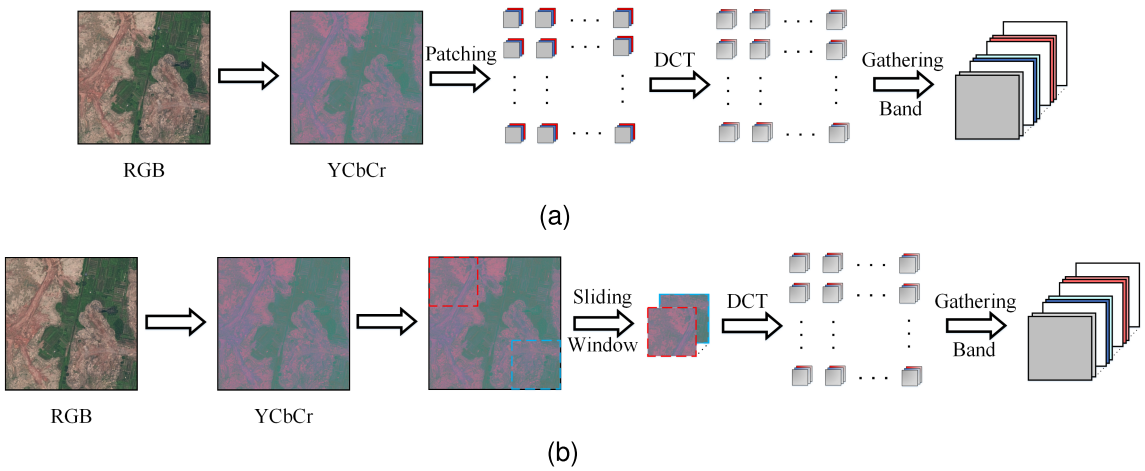


Fig. 3. (a) Conventional DCT-based methods apply DCT to fixed-size, nonoverlapping patches within each channel. (b) In contrast, our proposed approach performs DCT over sliding windows within each channel, allowing customized window size and stride. This flexibility enhances the suitability of the features for subsequent difference analysis. Notably, SDCT is applied directly to the features without conversion to the YCbCr domain.

$\{D_1, D_2\}$. Subsequently, the CDA generates the initial property difference descriptors $\{M_{1,2}, M_{2,1}\}$. These descriptors are then refined through layer normalization (LN) and a multilayer perceptron (MLP), resulting in the final property difference descriptors $\{D_{1,2}, D_{2,1}\}$, which represent the interobject comparisons between bitemporal images. The overall procedure of DA can be summarized as follows:

$$\begin{aligned} D_1 &= \mathrm{SDCT}(F_1) \\ D_2 &= \mathrm{SDCT}(F_2) \\ M_{1,2} &= \mathrm{CDA}(D_1, D_2) \\ M_{2,1} &= \mathrm{CDA}(D_2, D_1) \\ D_{1,2} &= \mathrm{MLP}\left(\mathrm{LN}\left(M_{1,2} + F_1\right)\right) + M_{1,2} \\ D_{2,1} &= \mathrm{MLP}\left(\mathrm{LN}\left(M_{2,1} + F_2\right)\right) + M_{2,1}. \end{aligned} \tag{2}$$

1) Sliding DCT: As shown in Fig. 3(a), the method [21] divides images into nonoverlapping patches of fixed size $P \times P$, performing DCT on each patch across all channels. The resulting frequency components from patches corresponding to the same frequency band are gathered and reconstructed into individual channels, preserving the original spatial alignment. Through this process, red, green, blue (RGB) images $\in \mathbb{R}^{3\times H\times W}$ are transformed into frequency feature maps $\in \mathbb{R}^{3\times P^2\times H/P\times W/P}$. However, this fixed patch-based design poses limitations, as it struggles to balance spatial resolution and band diversity. More critically, the rigid patch-splitting strategy significantly disrupts spatial continuity, which is detrimental to pixel-level classification tasks.

To address these issues, we propose the SDCT, which enables DCT using a sliding window with a customized size and stride, as shown in Fig. 3(b). In this article, SDCT is applied to features without converting them into the YCbCr color space. Given input features $\in \mathbb{R}^{c\times h\times w}$, SDCT produces property descriptors $\in \mathbb{R}^{c\times p\times p\times h/s\times w/s}$. The size of the window $p \times p$ correlates with the frequency band length, while the stride of the window s controls the spatial sizes of the descriptors. This customized framework relieves the model from the rigid constraints of traditional DCT-based methods. Following the transformation, frequency components are aggregated in a similar manner to [21]. After that, each channel of these descriptors corresponds to features within a specific band, thus representing distinct object properties. Specifically, the features from the lowest frequency band are assigned to the first channel of the descriptors, those from the second-lowest frequency band to the second channel, and so on. Since features across different frequency bands are distinct, we assume they represent different properties. Thus, each channel of our descriptors represents a specific property. For instance, channels corresponding to low-frequency features capture color-related properties, while those containing high-frequency features represent texture-related properties.

Although our modifications to the previous work are relatively minimal, they yield substantial performance gains. In addition to its enhanced flexibility, SDCT is better suited for pixel-level classification tasks. More importantly, it generates high-quality property descriptors, and its customized design allows for the exploration of optimal descriptor configurations across diverse RS scenarios. This establishes a solid foundation for effective subsequent difference analysis.

2) Cross-Descriptor Attention: As shown in Fig. 2(b), the CDA is designed to compare bitemporal property descriptors. It is important to note that frequency features derived via DCT tend to concentrate more energy in low-frequency bands. This imbalance causes low-frequency feature components to dominate predictions, potentially leading to biased outcomes and degraded performance. Furthermore, such uneven energy distribution hinders the model's ability to approximate human-like adaptive reasoning. To address this issue, we introduce an energy balancer, implemented by a batch normalization (BN) layer, to normalize feature energy across frequency bands (channels), ensuring equitable consideration of all properties in the difference analysis. Following this, a depthwise convolution (DWConv) is applied to further smooth the normalized

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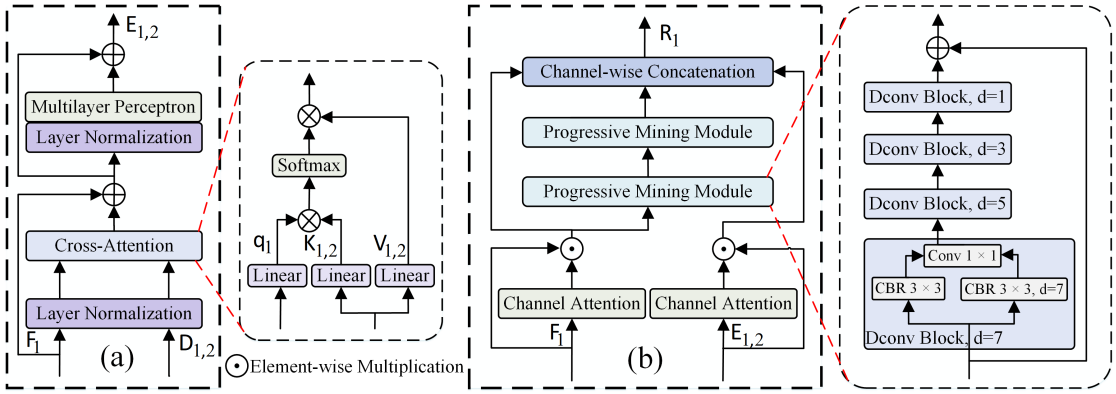


Fig. 4. (a) Illustration of the cross-transformer decoder, using the first instance as an example. For clarity, reshape operations are omitted, and a single-head cross-attention mechanism is depicted. (b) Illustration of the proposed RM module, also using the first instance as an example, where CBR denotes a convolutional layer followed by BN and ReLU activation, and d represents the dilation rate.

descriptors

$$D_1 = \mathrm{DWConv}_{7\times7}\left(\mathrm{BN}\left(D_1\right)\right)$$
$$D_2 = \mathrm{DWConv}_{7\times7}\left(\mathrm{BN}\left(D_2\right)\right). \tag{3}$$

Next, CDA adaptively selects and integrates the properties through learnable linear layers

$$q_1 = \mathrm{LN}\left(D_1\right)W_{q_1}, \quad k_1 = \mathrm{LN}\left(D_1\right)W_{k_1}$$
$$k_2 = \mathrm{LN}\left(D_2\right)W_{k_1}, \quad v_1 = \mathrm{LN}\left(D_1\right)W_{v_1} \tag{4}$$

where W_{q_1}, W_{k_1}, and W_{v_1} are linear layers. Technically, (4) enables the model to adaptively suppress interfering properties and activate discriminative ones through learnable weights. Subsequently, CDA compares the processed bitemporal property descriptors to produce the initial property difference descriptors $\{M_{1,2}, M_{2,1}\}$. For clarity, we illustrate the single-head version of CDA, using the computation of $M_{1,2}$ as an example

$$M_{1,2} = \mathrm{Softmax}\left(\frac{\left(q_1\left(k_1\right)^T - q_1\left(k_2\right)^T\right)^2}{\sqrt{d_1}}\right)v_1 \tag{5}$$

where d_1 is a scale factor.

To explain the comparison mechanism in (5), we consider two scenarios. In the case where no changes occur between the bitemporal images, the corresponding property descriptors remain similar across all properties. As a result, the inner product between q_1 and k_1 closely matches that between q_1 and k_2, yielding a negligible difference that contributes minimally to the property difference descriptors. Conversely, when changes are present, the inner product between q_1 and k_1 significantly differs from that between q_1 and k_2, effectively capturing meaningful differences from v_1 in the property difference descriptors.

To account for the sign of these inner product differences, we apply elementwise squaring to emphasize the magnitude of differences regardless of direction. Even in challenging cases, such as when certain properties of the same object differ, (5) remains effective, as (4) can suppress interfering properties, mitigating incorrect inner product differences. While the operational flow of our DA differs from the stepwise human difference analysis process, it offers a more streamlined, attention-based implementation without compromising effectiveness.

D. Cross-Transformer Decoder

The difference information extracted by the DA module is embedded in $D_{1,2}$ and $D_{2,1}$. However, due to the stride of SDCT being set to at least 2 to reduce computational cost, these descriptors are constrained by coarse-grained resolution. To address this, as shown in Fig. 4(a), we employ a cross-transformer decoder to upsample $D_{1,2}$ and $D_{2,1}$ to a finer resolution, resulting in enhanced features $E_{1,2}$ and $E_{2,1}$. For clarity, we illustrate the single-head version of the cross-transformer decoder using $E_{1,2}$ as an example

$$\begin{aligned} q_1 &= \mathrm{LN}\left(F_1\right)W_{q_{D1}} \\ k_{1,2} &= \mathrm{LN}\left(D_{1,2}\right)W_{k_{D1}} \\ v_{1,2} &= \mathrm{LN}\left(D_{1,2}\right)W_{v_{D1}} \end{aligned} \tag{6}$$

$$N_{1,2} = \mathrm{Softmax}\left(\frac{q_1\left(k_{1,2}\right)^T}{\sqrt{d_2}}\right)v_{1,2}$$
$$E_{1,2} = \mathrm{MLP}\left(\mathrm{LN}\left(N_{1,2} + F_1\right)\right) + N_{1,2} \tag{7}$$

where $W_{q_{D1}}, W_{k_{D1}}$, and $W_{v_{D1}}$ are linear layers, and d_2 denotes a scale factor.

E. Refinement Module

As illustrated in Fig. 4(b), the RM is proposed to enhance the spatial details of the input features. It begins by suppressing irrelevant information in F_1 and $E_{1,2}$ using channel attention [52]. To reduce model complexity, only F_1 is passed through the progressive mining module (PMM), which incrementally refines features from large to small receptive fields. This design is particularly beneficial in RS tasks involving targets of varying sizes. In addition, standard convolution is employed to maintain feature compactness. The procedure of the PMM can be described as follows:

$$\begin{aligned} \hat{X}^7 &= \mathrm{Conv1}\times1\left(\mathrm{Cat}\left(\mathrm{CBR}\left(\hat{X}\right), \mathrm{CBR}_{d=7}\left(\hat{X}\right)\right)\right) \\ \hat{X}^5 &= \mathrm{Conv1}\times1\left(\mathrm{Cat}\left(\mathrm{CBR}\left(\hat{X}^7\right), \mathrm{CBR}_{d=5}\left(\hat{X}^7\right)\right)\right) \\ \hat{X}^3 &= \mathrm{Conv1}\times1\left(\mathrm{Cat}\left(\mathrm{CBR}\left(\hat{X}^5\right), \mathrm{CBR}_{d=3}\left(\hat{X}^5\right)\right)\right) \\ \hat{X}^1 &= \mathrm{Conv1}\times1\left(\mathrm{Cat}\left(\mathrm{CBR}\left(\hat{X}^3\right), \mathrm{CBR}_{d=1}\left(\hat{X}^3\right)\right)\right) \end{aligned} \tag{8}$$

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where CBR(·) denotes a convolution block consisting of a 3×3 convolution, followed by BN and a rectified linear unit (ReLU) activation. The variable d indicates the dilation rate, $\hat{X}$ is the input feature, and $\hat{X}^1$ is the output feature generated by the PMM.

Furthermore, the operations of the RM can be detailed as follows (illustrated using the first one as an example):

$$\begin{aligned} F_1^c &= \mathrm{CA}\left(F_1\right) \cdot F_1 \\ E_{1,2}^c &= \mathrm{CA}\left(E_{1,2}\right) \cdot E_{1,2} \\ F_1^p &= \mathrm{PMM}\left(\mathrm{PMM}\left(F_1^c\right)\right) \\ R_1 &= \mathrm{Cat}\left(F_1^c, E_{1,2}^c, F_1^p\right) \end{aligned} \tag{9}$$

where CA(·) denotes the channel attention module as proposed in [52], and · indicates elementwise multiplication.

F. Loss Function

In RSCD tasks, changed pixels typically account for a much smaller proportion compared with unchanged pixels, leading to a significant class imbalance issue. To address this, we adopt a hybrid loss function that merges binary cross-entropy (BCE) loss with Dice loss [53]. The hybrid loss is formulated as follows:

$$L_{\text{hybrid}} = L_{\text{BCE}} + \lambda_0 L_{\text{Dice}} \tag{10}$$

where λ_0 is a hyperparameter that is set to 1. The detailed hyperparameter analysis is presented in Section IV-E3. The two individual loss functions can be formulated as

$$L_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N} y_i \log_2 \hat{y}_i + (1-y_i)\log_2(1-\hat{y}_i) \tag{11}$$

$$L_{\text{Dice}} = 1 - \frac{2\sum_{i=1}^{N} y_i \hat{y}_i}{\sum_{i=1}^{N} y_i + \hat{y}_i} \tag{12}$$

where $\hat{y}_i$ denotes the predicted change probability at the ith pixel of the change map, y_i is the corresponding ground truth label, and N represents the total number of pixels in the change map.

IV. Experiments

A. Datasets

To comprehensively evaluate the performance of the proposed LHICD-Net, we perform experiments on three representative RSCD datasets: the widely used LEVIR dataset [10], the challenging cropland change detection (CLCD) [54] dataset, and the season-varying change detection (CDD) dataset [55].

1) LEVIR: The LEVIR dataset comprises 637 pairs of RS images, each with a spatial resolution of 1024×1024 pixels, covering 20 regions primarily focused on building changes. To reduce computational overhead, the original images are cropped into patches of 256×256 pixels, resulting in training, validation, and test sets containing 7120, 1024, and 2048 image pairs, respectively.

2) CLCD: The CLCD dataset consists of 600 pairs of RS images, each with a resolution of 512×512 pixels, and focuses on multiclass CD involving objects, such as buildings, roads, lakes, and cropland. To reduce computational cost, the original images are cropped into 256×256 pixel patches, yielding 1440 images for training, 480 images for validation, and 480 images for testing.

3) CDD: The CDD dataset comprises 11 pairs of seasonally varying Google Earth images, targeting multiclass CD involving objects, such as buildings, roads, trees, and vehicles. To reduce computational complexity, the original images are divided into patches of 256×256 pixels. The dataset is partitioned into 10 000 training pairs, 3000 validation pairs, and 3000 testing pairs.

B. Implementation Details

The proposed LHICD-Net is implemented using an NVIDIA GeForce RTX 3090 GPU. Optimization is performed using stochastic gradient descent with an initial learning rate of 0.01, which decreases linearly over the course of training. The network is trained for a total of 300 epochs with a batch size of 12.

C. Evaluation Metrics

To quantitatively assess the performance of LHICD-Net, four widely used metrics are employed: precision (Pre), recall (Rec), intersection over union (IoU), and $F1$ score [48], [56], [57]. These evaluation metrics are formally defined as follows:

$$\text{Pre} = \frac{\text{Tp}}{\text{Tp}+\text{Fp}} \tag{13}$$

$$\text{Rec} = \frac{\text{Tp}}{\text{Tp}+\text{Fn}} \tag{14}$$

$$\text{IoU} = \frac{\text{Tp}}{\text{Tp}+\text{Fp}+\text{Fn}} \tag{15}$$

$$\text{F1} = \frac{2\text{Pre}\cdot\text{Rec}}{\text{Pre}+\text{Rec}} \tag{16}$$

where Tp, Tn, Fp, and Fn denote true positive, true negative, false positive, and false negative pixels, respectively.

D. Performance Comparison

To comprehensively evaluate the proposed method, we compare it with 12 representative methods, including: fully convolutional Siamese-difference (FC-Diff) [47], fully convolutional Siamese-concatenation (FC-Conc) [47], Changeformer [16], efficient local-global context aggregation network (ELGC) [11], Changer [17], bitemporal image transformer (BIT) [58], summator-subtractor network (SSCD) [12], cross-temporal difference transformer (CTD-Former) [19], spatiotemporal enhancement and interlevel fusion network (SEIF) [59], progressive refinement network (PRNet) [60], semantic-aware change detection network (SA-CDNet) [61], and deeply supervised hybrid feature aggregation network (DS-HyFA-Net) [51]. All comparison models are implemented using their default parameters as specified in the respective original publications. For fair comparison, the experimental conditions are maintained consistently with those described in Section IV-B.

1) Quantitative Comparison: Table I presents the quantitative comparison results across the evaluated methods. LHICD-Net consistently achieves the highest performance across all three datasets, significantly outperforming competing approaches in terms of both IoU and $F1$. Notably, on the widely used LEVIR dataset and the more challenging CLCD dataset, LHICD-Net records the top values for both Rec

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TABLE I

QUANTITATIVE COMPARISON RESULTS BASED ON IOU, $F1$, REC, AND PRE ACROSS THREE RSCD DATASETS. THE BEST AND SECOND-BEST PERFORMANCES ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY. ALL VALUES ARE PRESENTED AS PERCENTAGES (%). AN ASTERISK ("*") INDICATES RESULTS REPORTED IN THE ORIGINAL PAPER. THE PARAMETERS OF LHICD-NET ARE PRESENTED IN THE TABLE II

Method	Params (M)	LEVIR				CLCD				CDD			
		IoU	/ F1	/ Rec	/ Pre	IoU	/ F1	/ Rec	/ Pre	IoU	/ F1	/ Rec	/ Pre
FC-Diff	1.35	79.60	88.64	85.88	91.58	42.91	60.06	63.13	57.27	68.82	81.53	74.36	90.23
FC-Conc	1.54	79.99	88.88	86.83	91.04	42.05	59.20	63.28	55.62	68.47	81.29	73.98	90.20
Changeformer	41.05	80.99	89.50	87.94	91.11	48.88	65.66	65.24	66.09	91.16	95.37	95.21	95.54
ELGC	10.57	80.41	89.14	87.98	90.34	44.95	62.02	63.56	60.54	83.37	90.93	89.65	92.25
Changer	11.39	83.90	91.25	89.78	92.77	41.42	58.57	47.94	75.28	92.71	96.22	95.78	96.66
BIT	11.91	83.16	90.81	89.74	91.90	54.62	70.65	72.75	68.66	93.89	96.85	96.58	97.11
SSCD	12.23	82.71	90.54	89.29	91.83	55.09	71.05	68.34	73.98	95.16	97.52	97.31	97.74
CTD-Former	13.45	84.42	91.55	90.31	92.84	57.12	72.71	71.52	73.95	95.26	97.57	97.06	98.10
SEIF	28.60	82.87	90.63	89.94	91.34	55.73	71.57	70.98	72.18	94.03	96.92	96.51	97.34
PRNet	3.41	82.78	90.58	89.88	91.29	51.91	68.34	63.88	73.47	90.53	95.03	94.22	95.86
SA-CDNet	77.08	83.00	90.71	89.25	92.22	52.37	68.74	67.02	70.55	93.02	96.39	95.95	96.83
DS-HyFA-Net	11.08	82.51	90.41	88.91	91.97	55.34	71.25	71.76	70.75	93.42	96.60	96.07	97.14
LHICD-Net (ours)	-	85.27	92.05	91.27	92.84	61.21	75.94	76.55	75.34	95.45	97.67	97.12	98.23

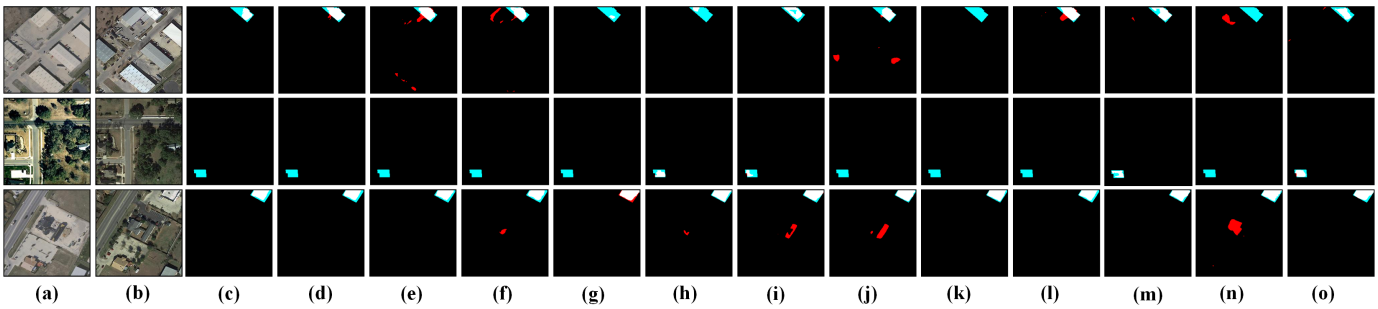


Fig. 5. Qualitative comparison between LHICD-Net and other methods on the LEVIR dataset. (a) T1 images. (b) T2 images. (c) FC-Diff. (d) FC-Conc. (e) ChangeFormer. (f) ELGC. (g) Changer. (h) BIT. (i) SSCD. (j) CTD-Former. (k) SEIF. (l) PRNet. (m) SA-CDNet. (n) DS-HyFA-Net. (o) Proposed method. The visualization uses white for true positives (Tp), red for false positives (Fp), black for true negatives (Tn), and blue for false negatives (Fn).

and Pre, highlighting its effectiveness in addressing pseudo and undetected changes. Specifically, the proposed LHICD-Net obtains IoU values of 85.27%, 61.21%, and 95.45%, and corresponding $F1$ scores of 92.05%, 75.94%, and 97.67% on the three public RSCD datasets. Compared with the second-best method, these results represent improvements of 0.85%, 4.09%, and 0.19% in IoU, and 0.50%, 3.23%, and 0.10% in $F1$ score, respectively. Furthermore, the Rec improvements on the LEVIR and CLCD datasets reach 0.96% and 3.80%, respectively, further demonstrating the effectiveness of LHICD-Net in addressing unidentified changes.

2) Qualitative Comparison: As illustrated in Figs. 5 and 6, LHICD-Net effectively eliminates pseudo changes introduced by variations in light-induced colors. In addition, it accurately detects subtle changes in buildings and cropland. In comparison with existing approaches, LHICD-Net exhibits enhanced robustness against both pseudo and undetected changes. Furthermore, the visual examples in Fig. 7 indicate that our predicted maps are more detailed and precise, demonstrating strong capability in preserving fine structures. This improvement is primarily due to the preservation of high-frequency information and the implementation of the refinement stage. Importantly, the fine-grained predictions show minimal sensitivity to high-frequency noise, which can be attributed to the use of learnable linear layers in our design.

E. Ablation Study

This section evaluates the effectiveness of the proposed modules and investigates the optimal configurations of SDCT across the three datasets. In addition, we provide an in-depth analysis of the DA mechanism by examining its property selection behavior. We also compare different descriptor generation methods on the challenging CLCD and LEVIR datasets. As indicated in Table II, the performance gains achieved by our modules on the CDD dataset are relatively limited, mainly due to its simplicity. Although the CDD dataset does not fully reflect the model's ability to handle complex scenarios,

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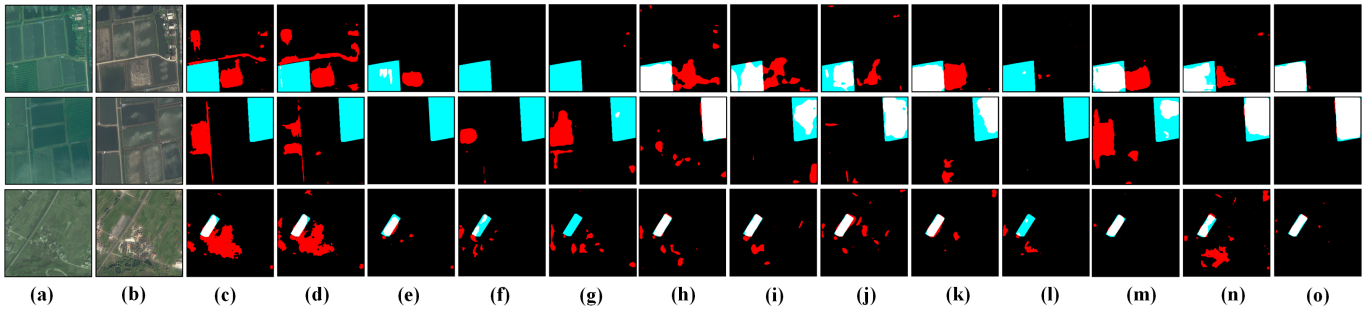


Fig. 6. Qualitative comparison between LHICD-Net and other methods on the CLCD dataset. (a) T1 images. (b) T2 images. (c) FC-Diff. (d) FC-Conc. (e) ChangeFormer. (f) ELGC. (g) Changer. (h) BIT. (i) SSCD. (j) CTD-Former. (k) SEIF. (l) PRNet. (m) SA-CDNet. (n) DS-HyFA-Net. (o) Proposed method. The visual results are color-coded as follows: white for true positives (Tp), red for false positives (Fp), black for true negatives (Tn), and blue for false negatives (Fn).

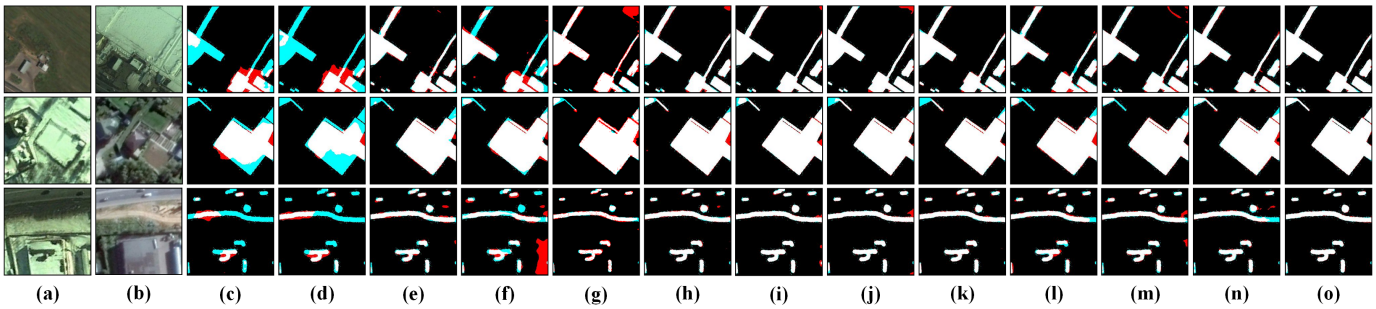


Fig. 7. Qualitative comparison between LHICD-Net and other methods on the CDD dataset. (a) T1 images. (b) T2 images. (c) FC-Diff. (d) FC-Conc. (e) ChangeFormer. (f) ELGC. (g) Changer. (h) BIT. (i) SSCD. (j) CTD-Former. (k) SEIF. (l) PRNet. (m) SA-CDNet. (n) DS-HyFA-Net. (o) Proposed method. The results are visualized with the following color scheme: white for true positives (Tp), red for false positives (Fp), black for true negatives (Tn), and blue for false negatives (Fn).

TABLE II

QUANTITATIVE EVALUATION OF LHICD-NET UNDER VARIOUS CONFIGURATIONS. RESULTS ON THE THREE DATASETS ARE REPORTED IN PERCENTAGE (%), WITH THE BEST OUTCOMES HIGHLIGHTED IN BOLD. "=" INDICATES NO CHANGE IN THE NUMBER OF PARAMETERS. $p \times p$ AND s REPRESENT THE SLIDING WINDOW SIZE AND STRIDE, RESPECTIVELY

No.	Setting	Params (M)	LEVIR	CLCD	CDD
			IoU / F1 / Rec / Pre	IoU / F1 / Rec / Pre	IoU / F1 / Rec / Pre
HRS					
#01	w/o HRS	-0.03	84.78 / 91.76 / 91.41 / 92.11	59.15 / 74.33 / 68.78 / **80.86**	95.34 / 97.61 / 97.01 / **98.23**
CDA					
#02	w/o CDA	=	84.86 / 91.81 / 90.90 / 92.74	57.76 / 73.22 / 75.44 / 71.13	95.35 / 97.62 / 97.03 / 98.21
SDCT					
#03	p=3, s=2	14.23	**85.27** / **92.05** / 91.27 / 92.84	58.25 / 73.62 / 70.11 / 77.48	95.33 / 97.61 / 97.05 / 98.17
#04	p=5, s=2	15.54	84.97 / 91.88 / 91.04 / 92.72	60.52 / 75.40 / 74.11 / 76.74	95.33 / 97.61 / 97.06 / 98.17
#05	p=5, s=4	15.54	84.42 / 91.55 / 90.87 / 92.24	60.75 / 75.59 / 71.59 / 80.06	95.33 / 97.61 / **97.12** / 98.10
#06	p=7, s=2	17.51	84.87 / 91.81 / 90.58 / **93.08**	58.86 / 74.10 / 71.28 / 77.15	**95.45** / **97.67** / **97.12** / **98.23**
#07	p=7, s=4	17.51	84.61 / 91.66 / **91.88** / 91.44	**61.21** / **75.94** / **76.55** / 75.34	95.29 / 97.59 / 96.96 / 98.22
RM					
#08	w/o RM	-0.82	84.55 / 91.63 / 90.96 / 92.31	59.28 / 74.44 / 72.63 / 76.34	95.06 / 97.46 / 96.85 / 98.09

it remains a representative and reliable benchmark for assessing RSCD model performance. Unless stated otherwise for SDCT-specific analysis, all experiments are conducted using the respective optimal SDCT configurations.

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Fig. 8. Grad-CAM visualization results illustrating feature changes before and after processing by DA. (a) T1 image, (b) T1 features prior to DA, (c) T1 features following DA, (d) T2 image, (e) T2 features prior to DA, (f) T2 features following DA, and (g) ground truth. Red denotes regions with closer attention, while blue indicates lower attention.

1) Effectiveness of the HRS: The HRS module is integrated into the encoder to mitigate high-frequency information loss during downsampling, thereby preserving band diversity in the property descriptors and better aligning with human perception across spectral bands. To assess its contribution, we perform an ablation study by excluding HRS (denoted as (#01). As reported in Table II, the $F1$ scores for #01 decline by 0.29%, 1.61%, and 0.06% across the three datasets, respectively. These findings indicate that the loss of high-frequency properties can lead to inaccurate CD and confirm the effectiveness of the HRS module.

2) Effectiveness of the DA: Inspired by the human difference analysis strategy, DA is specifically designed to implement this approach. To better evaluate the effectiveness of DA, we apply Grad-CAM [62] to visualize the features before and after processing through DA. As illustrated in Fig. 8, the main region of identical buildings with interfering color variations is suppressed after DA processing, while greater attention is accurately focused on the key property of buildings, namely their boundaries. We argue that this attention selection strategy aligns with the human difference analysis strategy. Consequently, the visual results indicate that DA effectively implements this strategy. To further assess the superiority of CDA, we replace CDA with standard cross-attention (#02). As reported in Table II, the $F1$ scores of #02 decline by 0.24%, 2.72%, and 0.05% on the three datasets, respectively. These results demonstrate that LHICD-Net integrated with CDA is more competitive.

3) Hyperparameter Analysis of the Loss Function: In our method, we introduce a hyperparameter λ_0 to balance the BCE loss and Dice loss within the hybrid loss function. To investigate the impact of different λ_0 values, we evaluate five values of λ_0 through experiments on three datasets. As shown in Table III, the best performance is achieved when λ_0 is set to 1 across all three datasets, indicating that BCE loss and Dice loss contribute equally to the optimization. Notably, setting λ_0 to 0.8 also achieves the best result on the CDD dataset. Moreover, we observe that the performance on CDD is only marginally affected by different values of λ_0, indicating that the model is relatively insensitive to the choice of λ_0 on this dataset. Considering that the value of λ_0 has a significant impact on the performance of both LEVIR and CLCD datasets, we ultimately set λ_0 to 1 for consistently superior performance.

4) Optimal Settings of SDCT: The RSCD datasets contain various types of changed objects, which leads to different band requirements for accurate comparison. In addition, objects of different scales may be more effectively captured by descriptors of particular sizes. Therefore, we investigate the optimal SDCT settings on the three datasets, as shown in Table II. On the LEVIR and CDD datasets, which contain many small-scale objects, our LHICD-Net achieves significantly improved detection accuracy when $s = 2$ is used. This configuration helps retain fine-grained spatial information, which is essential for detecting small-scale objects. In contrast, the CLCD dataset contains large and regular targets, such as cropland. Thus, setting $s = 4$ facilitates effective modeling of region-level semantics. For the CDD dataset, which includes diverse changed objects, such as buildings, roads, and vegetation, setting $p = 7$ is more effective because more varying properties can be utilized to compare diverse objects. For the cropland-dominant CLCD dataset, setting $p = 7$ achieves the best performance as more high-frequency properties (e.g., texture) are crucial for identifying changed cropland. However, for the LEVIR dataset, which focuses solely on building CD, $p = 3$ is sufficient and prevents interference from unrelated properties. Compared with the worst-performing settings, the optimal settings improve $F1$ scores by 0.50%, 2.32%, and 0.08% on the three datasets, indicating the value of the exploration. Notably, changing the sliding window size naturally impacts the number of parameters. Fortunately, the parameters in our method remain within a reasonable range compared with the methods in Table I.

TABLE III
QUANTITATIVE COMPARISONS OF THE PROPOSED METHOD WITH DIVERSE HYPERPARAMETER SETTINGS ON THREE DATASETS. THE MODIFIED HYPERPARAMETER SETTINGS ARE MARKED IN BOLD AND THE BEST RESULTS ARE HIGHLIGHTED IN RED

λ_0	LEVIR	CLCD	CDD
	IoU (%) / F1 (%)	IoU (%) / F1 (%)	IoU (%) / F1 (%)
0.6	83.72 / 91.14	59.40 / 74.53	95.42 / 97.66
0.8	84.83 / 91.79	60.71 / 75.55	95.45 / 97.67
1	85.27 / 92.05	61.21 / 75.94	95.45 / 97.67
1.2	84.87 / 91.82	61.11 / 75.86	95.39 / 97.64
1.4	84.53 / 91.62	60.13 / 75.10	95.34 / 97.62

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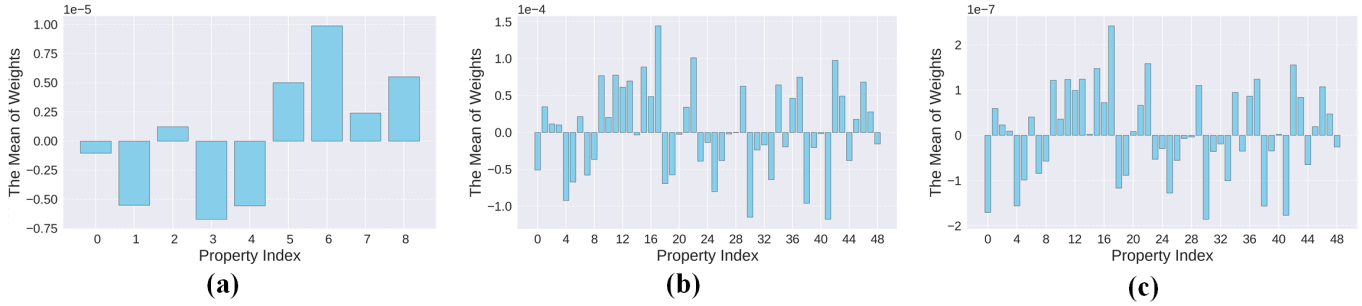


Fig. 9. Analysis of property selection under optimal SDCT configurations across three RSCD datasets. As the index rises, the corresponding property shifts from low-frequency to high-frequency components. (a) LEVIR dataset. (b) CLCD dataset. (c) CDD dataset.

5) Analysis of Property Selection: To further assess the working mechanism of our DA and the importance of the properties, we analyze the property selection across the three RSCD datasets using the optimal SDCT configuration. This analysis is grounded in three key observations from prior knowledge: 1) the weights of the linear layer associated with a particular channel can indicate the selection for that channel; 2) in RSCD, single-channel prediction maps are passed through a sigmoid function, yielding output values between 0 and 1, where 1 denotes change, and 0 denotes no change. Thus, positive weights typically indicate feature activation, while negative weights suggest suppression; and 3) in our approach, each channel of the property descriptor corresponds to a specific property. As shown in Fig. 9, we visualize property selection results on the three RSCD datasets by computing the mean weights [W_{q1}, W_{k1}, and W_{v1} in (4)] for each property (channel). On the LEVIR dataset, low-frequency properties tend to be suppressed while high-frequency properties are activated, which aligns with the human difference analysis strategy applied to this dataset. On the cropland-dominant CLCD dataset, the property selection is more complex. Although we cannot summarize a clear rule, the results still sufficiently indicate which properties are important (i.e., those properties whose corresponding weight averages are significantly greater than zero). A notable consistency in property selection is observed between the CLCD and CDD datasets, which will be further investigated in future work.

6) Effectiveness of the RM: The RM is introduced to enhance spatial details in the input features by extracting fine-grained information from multiple receptive fields. To assess its effectiveness, an ablation study is performed by removing RM (Configuration #08). As indicated in Table II, incorporating the RM improves the $F1$ scores by 0.42%, 1.50%, and 0.21% on the three datasets. These results demonstrate the effectiveness of the RM.

7) Discussion on the Descriptor Generation Methods: High-quality decomposition of object properties is essential for implementing the human difference analysis strategy. Accordingly, various descriptor generation methods are examined and evaluated for their effectiveness in addressing challenges on the CLCD and LEVIR datasets. Specifically, the methods include spatial–conv-based (using convolution with kernel sizes and strides matching the optimal SDCT configurations on the respective datasets), DWT-based, C-DCTn-based (using conventional DCT with patch size $n \times n$) methods, and the proposed method. As presented in Table IV, on the CLCD dataset, the spatial–conv-based method ranks as the second lowest in performance, while on the LEVIR dataset, it performs the worst. In contrast, most frequency-based methods demonstrate superior results, suggesting that comparisons in the spatial domain may be suboptimal. This outcome is attributed in part to the alignment of frequency-based methods with the human strategy of property-based comparison, which helps reduce the impact of interfering properties present in spatial–conv-based methods. However, frequency-based approaches often struggle to balance band diversity and spatial resolution. In addition, the performance of C-DCTn-based methods is hindered by the use of a rigid partitioning strategy. By contrast, the proposed method addresses these limitations and achieves the highest performance by leveraging appropriately selected frequency bands and consistent spatial representations.

TABLE IV

PERFORMANCE COMPARISON OF VARIOUS DESCRIPTOR GENERATION METHODS ON THE CLCD AND LEVIR DATASETS. THE HIGHEST AND SECOND-HIGHEST SCORES ARE MARKED IN RED AND BLUE, RESPECTIVELY

Method	CLCD	LEVIR
	IoU (%) / F1 (%)	IoU (%) / F1 (%)
Spatial-conv-based	56.69 / 72.36	84.38 / 91.53
DWT-based	58.18 / 73.56	84.58 / 91.65
C-DCT2-based	56.63 / 72.31	84.74 / 91.74
C-DCT4-based	60.01 / 75.01	84.74 / 91.74
C-DCT8-based	57.25 / 72.81	84.49 / 91.59
SDCT-based (ours)	61.21 / 75.94	85.27 / 92.05

To further assess the effectiveness of these descriptor generation methods, visual results on the CLCD dataset are provided in Fig. 10. In the first row, both the spatial–conv-based and C-DCT2-based methods fail to detect changed cropland with distinct textures. Increasing the number of available frequency bands improves performance in the C-DCT4-based method and resolves the issue in the C-DCT8-based method and the proposed approach. Furthermore, the proposed method is more effective in removing pseudo changes. These results validate the utility of property-based comparison and support

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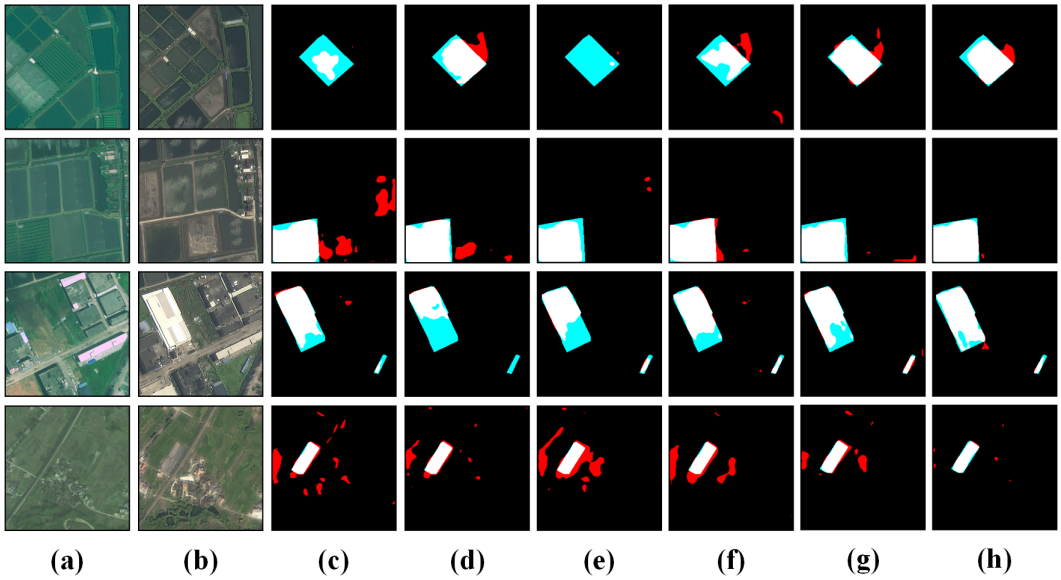


Fig. 10. Visual comparison of various descriptor generation approaches on the CLCD dataset. (a) T1 images. (b) T2 images. (c) Spatial–conv-based method. (d) DWT-based method. (e) C-DCT2-based method. (f) C-DCT4-based method. (g) C-DCT8-based method. (h) Proposed method.

the notion that changed objects typically possess distinctive discriminatory bands (properties).

V. Conclusion

Motivated by the effectiveness of the human difference analysis strategy in RSCD, which relies on property-based comparison, we introduce the DA to emulate this approach and move toward achieving human-level performance in RSCD. DA generates property descriptors, adaptively selects relevant properties, and derives interobject comparisons based on the selected property comparisons. To effectively incorporate DA into a deep learning framework, we propose LHICD-Net, which is an end-to-end network that systematically implements the four stages of the human RSCD process. The encoder, enhanced with an HRS, serves as the perception stage, which preserves high-frequency properties to align with human perception across diverse bands. DA functions as the analysis stage. The cross-transformer decoder and RM constitute the refinement stage by enhancing spatial details through the extraction of fine-grained features from various receptive fields. Finally, the predictor carries out the prediction stage.

In addition, we provide a deeper insight into the internal mechanism of our DA, confirming its consistency with the human difference analysis strategy through investigation of the property selection process. The effectiveness of the property-based comparison is further validated by evaluating the methods used for generating property descriptors. Importantly, LHICD-Net goes beyond superficial imitation, effectively addressing the two core challenges in RSCD and achieving state-of-the-art performance across three benchmark RSCD datasets.

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